Assignment 3

CONTAMINANT TRANSPORT TO A RIVERBANK FILTRATION SYSTEM

Group 19

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12104 Environmental Modelling

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Part 1: Groundwater flow

Task 1 and 2: Governing equation and model domain sketch

Since the model is assumed to be steady-state horizontal flow in a confined aquifer, and not includes infiltration and recharge, the divergence of mass flux is zero:

$$\nabla \cdot (-B \times K \times \nabla h) = 0$$

Where，

B is the thickness of confined aquifer, B=5 m;

K is the horizontal hydraulic conductivity, $K_{h1} = 5 \times \frac{10^{-3}m}{s}$, $K_{h2} = 1 \times \frac{10^{-3}m}{s}$;

∇h is the hydraulic gradient.

Top view can be used to model because the aquifer thickness is much smaller than the horizontal dimensions, and the hydraulic characteristics (K and porosity) remain approximately unchanged in the vertical dimension:

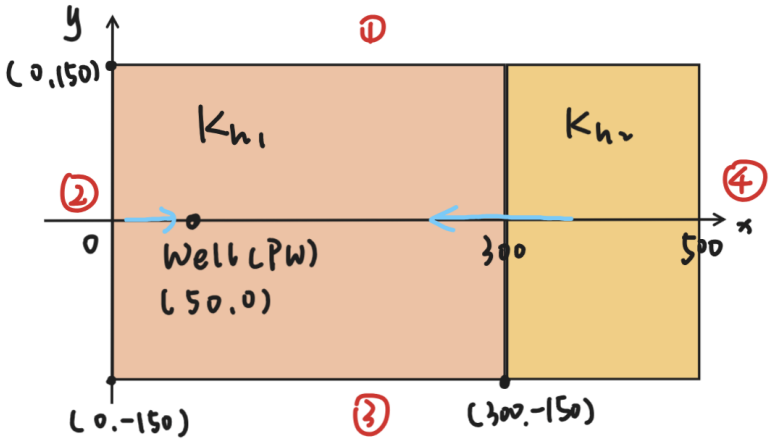


Figure 1 Model domain sketch of top view

Boundary Conditions are as follows：

Location	Boundary Type	Boundary condition
Side boundary (y=150)	Neumann	Zero flux
River (x = 0)	Dirichlet	H_{river}=12 m
Side boundary (y=-150)	Neumann	Zero flux
Right boundary (x = 500)	Dirichlet	H_{right}=12.5 m

Task 3: 2-D steady flow model under natural conditions

Without well pumping, groundwater flows uniformly from right to left under a constant horizontal hydraulic gradient, forming a one-dimensional steady flow field:

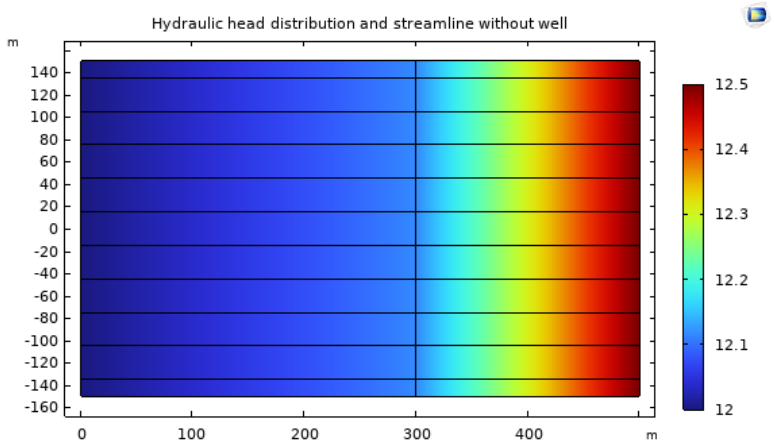


Figure 2 Hydraulic head distribution and flow field without well pumping

Task 4: Drawdown analysis for pumping scenarios

According to the model results, the average hydraulic head at the well location is 12.019 m in the absence of pumping. When the pumping rates are Q_1=1100 m^3/d and Q_2=180 m^3/h, the average hydraulic heads decrease to 11.425 m and 9.686 m, corresponding to drawdowns of 0.59 m and 2.33 m, respectively.

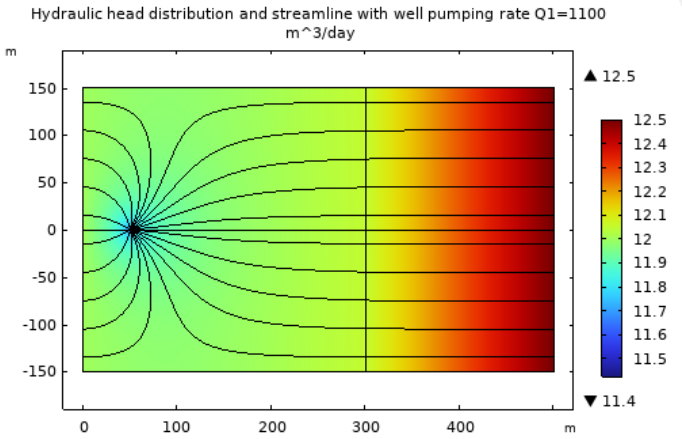


Figure 3 Hydraulic head distribution and flow field (Q_1=1100 m^3/d)

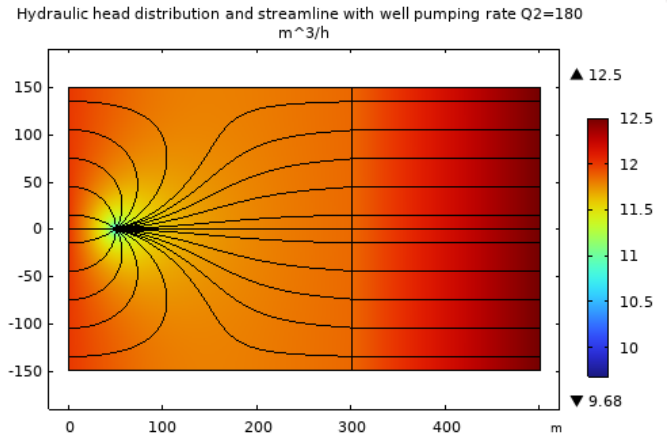
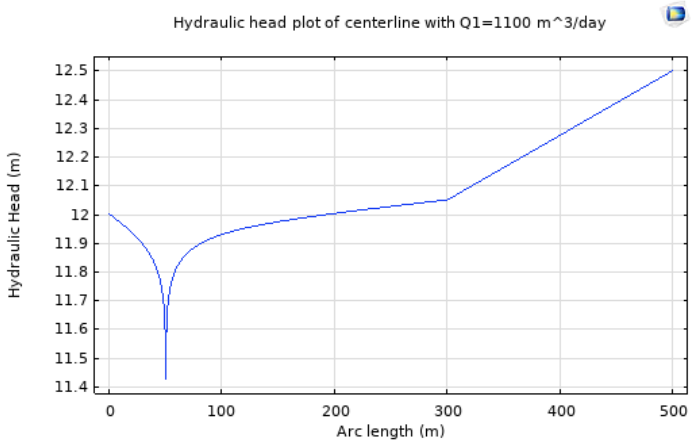


Figure 4 Hydraulic head distribution and flow field (Q_2=180 m^3/h)



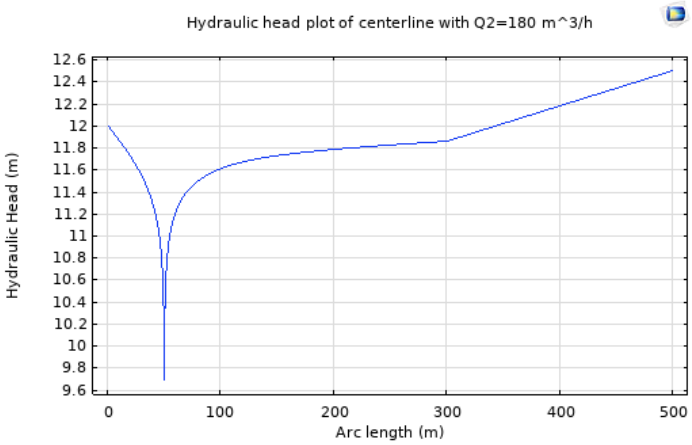


Figure 5 Hydraulic head changes along centerline (Q_1=1100 m^3/d)

Figure 6 Hydraulic head changes along centerline (Q_2=180 m^3/h)

Task 5: Estimation of groundwater travel time

When the pumping rate is 1100 m^3/day, the shortest travel time of water from the model boundary to the well is approximately 0.5 years, while the longest reaches 1.92 years. When the pumping rate increases to 110 m^3/h, the shortest travel time is around 0.2 years, and the longest about 1.03 years.

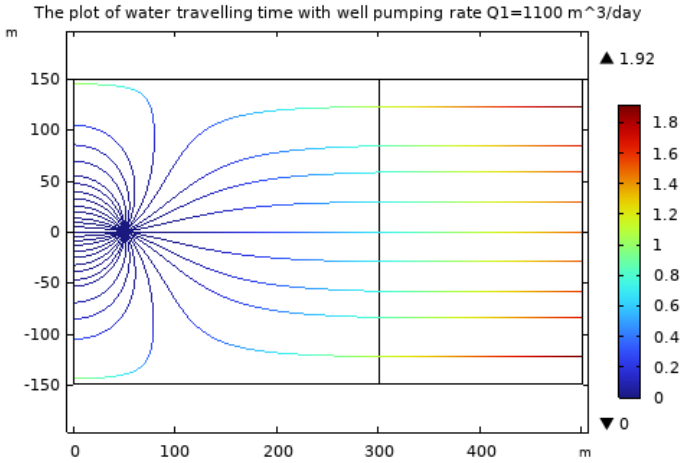


Figure 7 Water travelling time of particles (Q_1=1100 m^3/d)

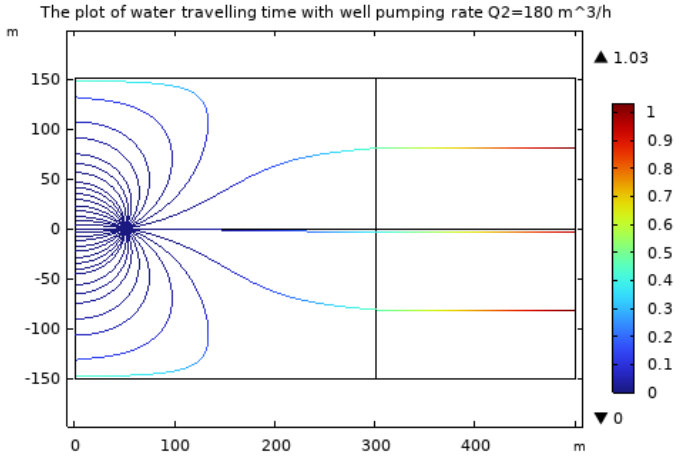


Figure 8 Water travelling time of particles (Q_2=180 m^3/h)

Task 6: Effect of aquifer thickness on drawdown behavior

Compared with Figure 3, when the aquifer thickness increases from 5 m to 10 m, the drawdown caused by well pumping decreases from 0.6 m to 0.3 m. According to the Theim equation, the hydraulic conductivity in this case is twice the original value, resulting in a drawdown that is reduced to half under the same pumping rate:

$$s = \frac{Q}{2\pi T} ln \frac{r_2}{r_1}$$

Where,

s is drawdown (m);

Q is water pumping volume (m^3/s);

T is transmissivity of the aquifer, T=K*B (m^2/s);

R_1, r_2 is radius of the inner and outer boundary of the well (m).

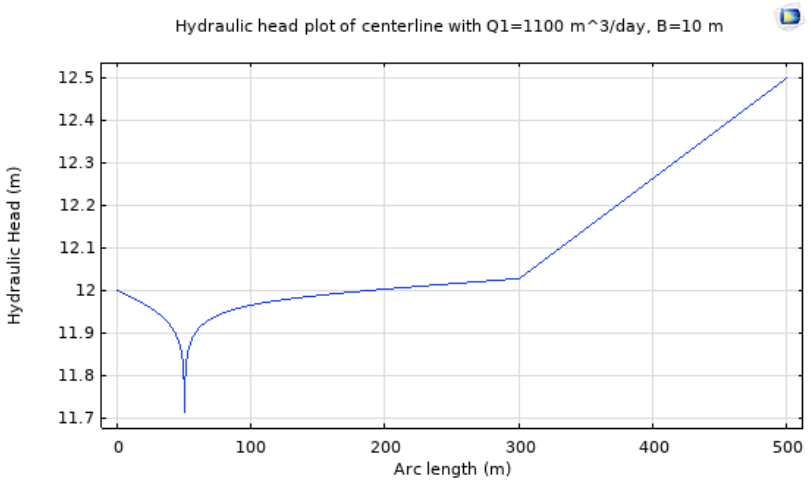


Figure 9 Hydraulic head changes along centerline (B=10 m, Q_1=1100 m^3/d)

Task 7: Flux partitioning between river and boundary inflows

The flux balance analysis using boundary probes reveals that the river is the dominant source of recharge to the pumping well under both pumping scenarios. For the lower pumping rate of 1100 m^3/d, approximately 73.54% of the extracted water originates from the river, while 26.5% is supplied from the right boundary. When the pumping rate is increased to 180 m^3/h, the river contribution increases markedly to about 90%, with only 10% coming from the right boundary.

Table 1 Simulated water fluxes and contributions from the river and the right boundary under two pumping rates

Pumping rate	Inflow from river	Contribution from river (%)	Inflow from right boundary	Contribution from right boundary (%)
1100 m^3/d	808.98 m^3/d	73.54	291.54 m^3/d	26.50
180 m^3/h	162.68 m^3/h	90.38	17.31 m^3/h	9.62

Task 8: Alternative 2-D model setup for riverbank filtration

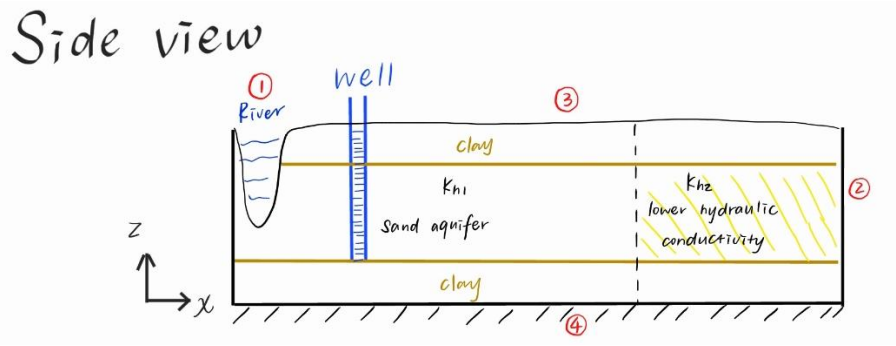


Figure 10 Side-view 2-D model setup for riverbank filtration

In addition to the horizontal (plan-view) 2-D model used in this assignment, the RBF system can also be represented by a vertical 2-D cross-section (x–z plane). This alternative setup focuses on vertical infiltration from the river through the riverbed into the aquifer and groundwater flow toward a pumping well.

This 2-D side-view setup can be used to: Quantify vertical infiltration and leakage through the riverbed layer;

Assess travel times of river water reaching the pumping well; Examine layered or anisotropic aquifers (e.g. $K_{h1} > K_{h2}$); Evaluate time-related capture zones.

Boundary Conditions：

Location	Boundary Type	Boundary condition
River (x = 0)	Dirichlet	H_{river}=12 m
Right boundary (x = L)	Dirichlet	H_{right}=12.5 m
Ground surface (z = z_s)	Neumann	Zero flux
Bottom (z = 0)	Neumann	Zero flux

Task 9: Mesh refinement strategy and justification

For accurate numerical results and stable particle-tracking performance, mesh refinement should focus on locations where hydraulic gradients or fluxes change sharply.

So the mesh should be locally refined in the following regions:

①. Near the well: The pumping well produces very steep hydraulic-head gradients. A fine mesh is required to resolve the drawdown cone and capture accurate velocities for travel-time estimation.

②. At the river boundary: The river-aquifer interface is the main inflow boundary. Refinement ensures accurate calculation of flux exchange and boundary probes used for river-water contribution.

③. At the interface between K_{h1} and K_{h2}: The contrast in hydraulic conductivity causes a discontinuity in flow direction. Refinement prevents spurious gradients and maintains continuity of flow lines.

④. Near the stagnation point: Fine mesh prevents particle stagnation errors.

Part 2: Contaminant transport

Task 10: Define the governing transport equation

We defined the governing transport equation for the 2-D top view setup with transient reactive transport, linear sorption and first-order degradation:

$$\phi R\frac{\partial c}{\partial t} + \nabla \cdot (\phi c v) - \nabla \cdot (\phi D \nabla c) + \phi k c = \frac{Q_m}{b}$$

$$\mathrm{R} = \left(1 + \frac{\rho_b}{\phi}\mathrm{K_d}\right)$$

Table 2 All variables of the governing transport equation

Variables	Parameter	Unit
c	Solute concentration	kg/m^3
ϕ	Porosity	\
R	Retardation factor	\
v	Pore velocity	m/d
D	Hydrodynamic dispersion tensor	m^2/d
Q_m	Source or sink term per unit volume	$kg \cdot m^{-3} \cdot d^{-1}$
k	First-order degradation rate constant	d^{-1}
b	Constant and uniform thickness	m
ρ_b	Bulk density	kg/L

Variables	Parameter	Unit
K_d	Sorption coefficient	L/kg

The mass sources and sinks Q_m are zero in this example.

The 2-D hydrodynamic dispersion tensor for this system is a symmetric 2×2 matrix, accounting for both mechanical dispersion and molecular diffusion coupled with transport porosity. For general flow conditions (not aligned with coordinate axes), it is defined based on the coupled parameters as:

$$D = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix}$$

To implement this, use the following defined expressions:

Table 3 2-D hydrodynamic dispersion tensor

Variables	Expression	Units
Dxx	$D_{xx} = D_{\text{eff}} + \alpha_T \lvert v \rvert + (\alpha_L - \alpha_T) \frac{v_x^2}{\lvert v \rvert}$	m^2/s
Dyy	$D_{xy} = D_{\text{eff}} + \alpha_T \lvert v \rvert + (\alpha_L - \alpha_T) \frac{v_y^2}{\lvert v \rvert}$	m^2/s
Dxy	$D_{xx} = (\alpha_L - \alpha_T) \frac{v_x}{\lvert v \rvert} v_y$	m^2/s
Deff	$D_{\text{eff}} = D_w \tau$	m^2/s

Task 11: Model setup: boundary conditions and source-sink representation

The initial condition assumes that at the start of the simulation(t=0), he contaminant concentration throughout the entire aquifer system is zero. The top and bottom boundaries are set as zero-flux boundaries, allowing no mass transfer across them. The left boundary(x=0) acts as a contaminant source and is assigned a Dirichlet boundary condition with a constant input concentration of C_{in}=1500 ng/L, while the right boundary represents clean water and is assigned a Dirichlet boundary condition with a concentration of zero. The pumping well is modeled as a flux/source, with the source term defined by $g = \frac{Q\,C}{2\pi r B}$, where Q is the pumping rate, C is the concentration at the well, r is the radial distance from the well, and B is the aquifer thickness. At the water supply well, the concentration must be specified according to the pumping rate and the local concentration, as the well extracts water at the concentrations determined by the transport model.

Task 12: Solute transport simulation under different pumping rates

Figures 11(a) and 11(b) show the concentration distributions for pumping rates Q_1 and Q_2. For Q_1, the high-concentration zone extends from the river toward the well in a wedge-like pattern, with moderate gradients and noticeable lateral spreading; the well acts as a concentration sink, lowering tracer levels nearby. For Q_2, the high-concentration region is confined near the river, with very steep gradients and minimal tracer near the well. The comparison highlights that higher pumping rates enhance advection, concentrating tracer near the well quickly while limiting lateral dispersion, whereas moderate pumping allows a more balanced advection–dispersion transport.

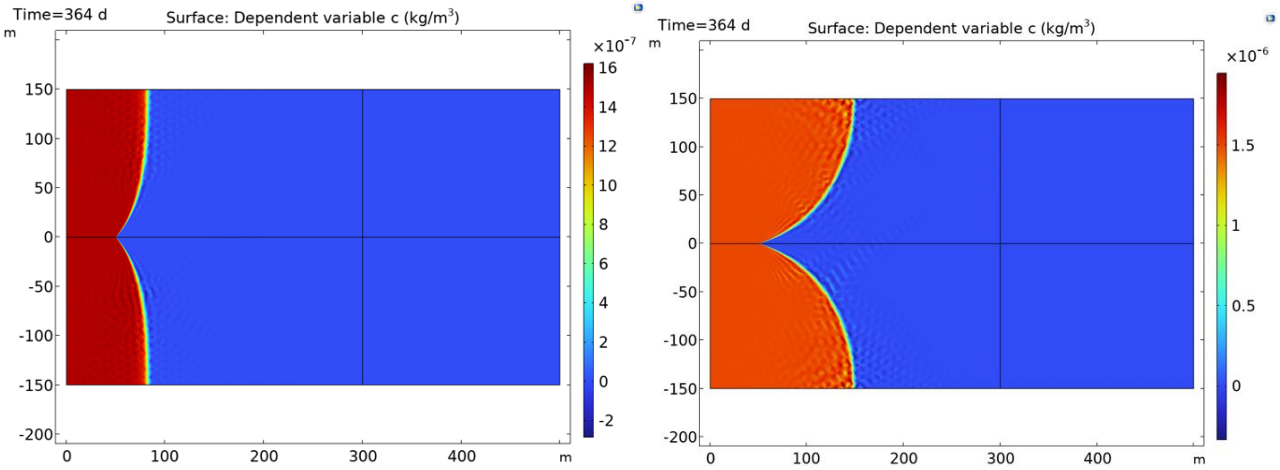


Figure 11 Concentration distribution after one year (a) Q_1=1100m^3/d; (b) Q_2=180m^3/h

Figures12(a) and 12(b) present the flow field and concentration distribution for the lowest pumping rate, Q_3. The flow is slow, with most groundwater moving toward the well and the river contributing little. The concentration distribution is smooth and widespread, with high tracer levels persisting near the well and extended low-concentration regions across the domain. Compared to Q_1 and Q_2, the Q_3 scenario illustrates the dominance of dispersion under low pumping, producing broader tracer spreading and weaker well "sink" effects.

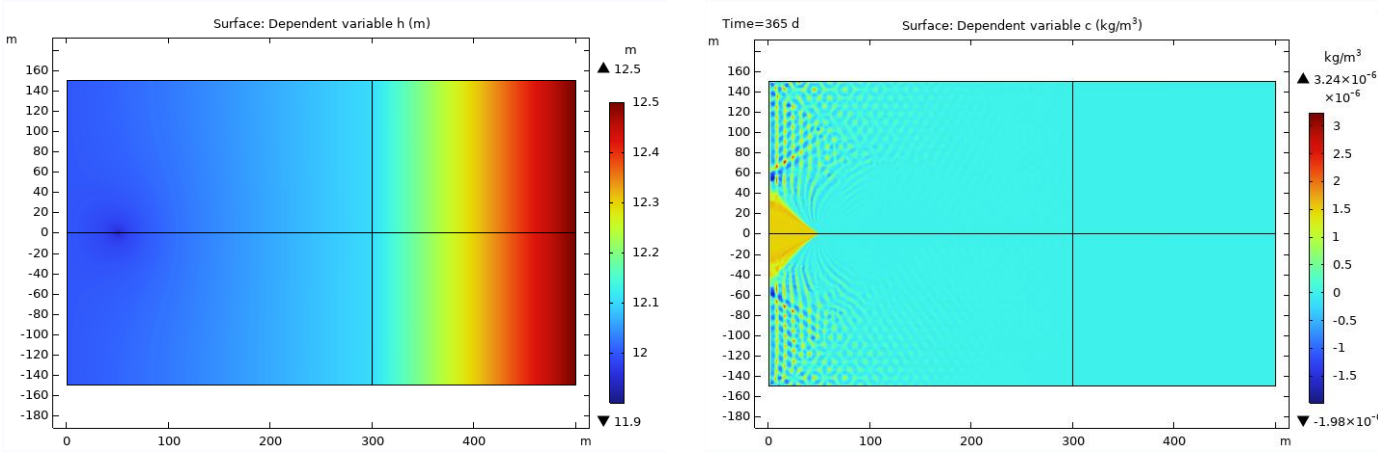


Figure 12 (a) flow field for Q_3=220m^3/d; (b) concentration distribution for Q_3=220m^3/d

Task 13: Temporal evolution of contaminant concentration at the pumping well

Figure 13 and 14 shows the average contaminant concentration at the pumping well for the three pumping rates over one year. For Q_1 (moderate pumping), concentration gradually rises and reaches an intermediate stable level. Q_2 (high pumping) increases fastest and achieves the highest stable concentration, reflecting rapid advection of the tracer. In contrast, Q_3 (low pumping) rises slowly and stabilizes at the lowest level, indicating dispersion-dominated transport under weak pumping.

The breakthrough curves highlight the effect of pumping rate on tracer transport. Higher pumping rates lead to earlier arrival and higher stable concentrations at the well, while lower rates delay arrival and reduce concentrations. This illustrates the balance between advection and dispersion: strong pumping accelerates tracer transport, whereas weak pumping allows more time for spreading, slowing accumulation.

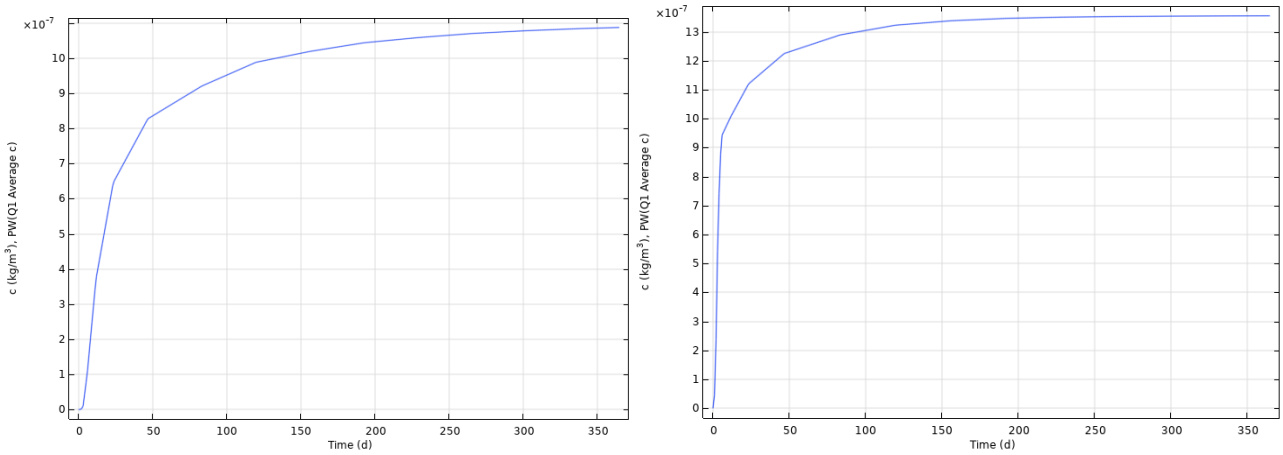


Figure 13 Average contaminant concentration at PW over time (a) Q_1=1100m^3/d; (b) Q_2=220 m^3/h

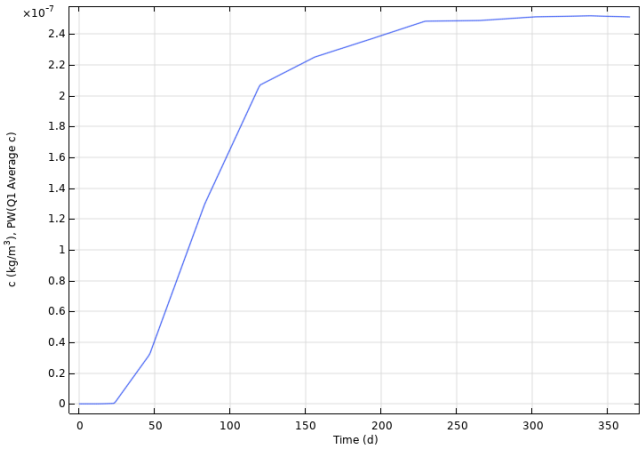


Figure 14 Average contaminant concentration at PW over time for Q_3=220m^3/d

Task 14: Analysis of Breakthrough Time and Hydraulic Travel Time

According to the results calculated from Python, the breakthrough curves show that under the two pumping rates, the times when the concentration at the pumping well reaches 50% of its maximum value are 19 days for Q_1 and 4 days for Q_2. The two corresponding breakthrough curves are shown in the figures below.

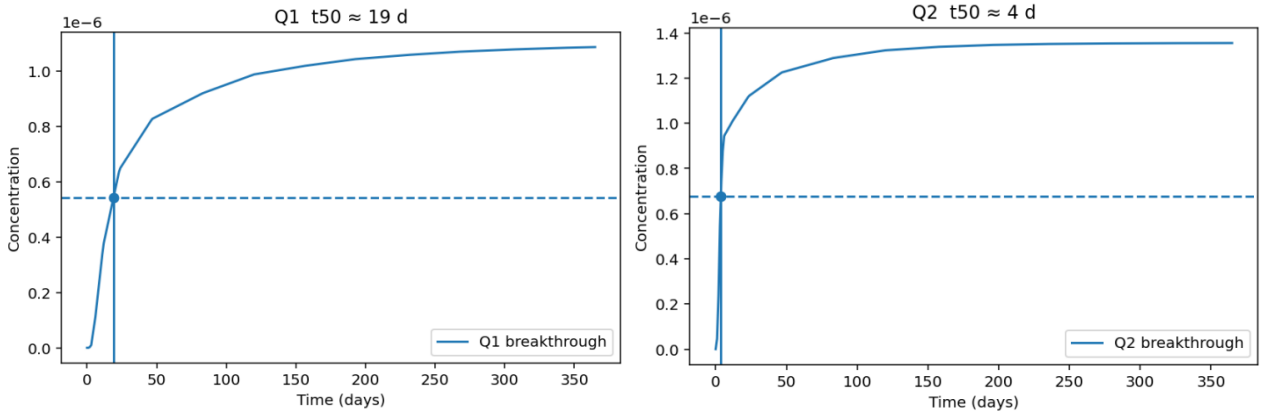


Figure 15 Time to 50 % breakthrough concentration (t_{50}) at the pumping well under Q_1 and Q_2

In comparison, the maximum groundwater travel times obtained in Question 5 were 1.92 days for Q_1 and 1.03

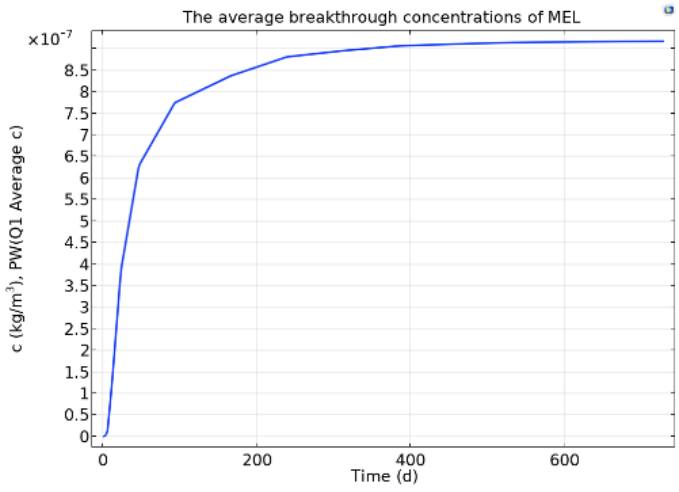
days for Q_2. The t_{50} values are significantly longer than the hydraulic travel times.
This indicates that it takes more time for the concentration to reach half of its maximum value than for the groundwater itself to reach the well along the shortest flow path. (Because of dispersion, mixing, and gradual replacement of clean water in the well.) The higher pumping rate (Q_2) produces stronger hydraulic gradients and faster flow velocities, which shorten both the hydraulic travel time and the t_{50} value. Hence, contaminants reach the well significantly earlier under Q_2 compared to Q_1.

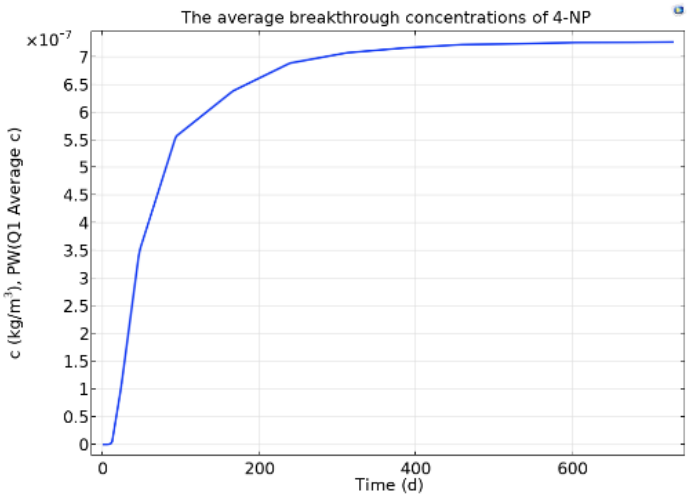
Task 15: Calculation of Degradation Rates for PMT Substances

The "persistence (days)" given in the Figure 3 is treated as the half-life ($t_{1/2}$) of a first-order decay process. According to the equation: $k = \frac{t_{1/2}}{ln2}$: k_{MEL}= 0.00578 d^{-1}; $k_{4\text{-}NP}$=0.01733 d^{-1}; k_{CBZ}=0.

Task 16: Simulation of Breakthrough Curves for PMT Substances

Compared with bromide, all three PMT compounds arrive later and reach lower concentrations at the pumping well. This is consistent with their larger retardation factors $R = 1 + \rho_b K_d/\phi$ and first-order degradation. Among the PMTs, the compound with the smallest K_d and/or lowest k (MEL) shows the earliest rise and the highest peak, the compound with the largest K_d and/or highest k (CBZ) exhibits the latest and smallest breakthrough.
Maximum concentrations at PW (over 2 years, Q_1): c_{MELmax}= 9.1729E-7; $c_{4\text{-}NPmax}$=7.2631E-7; c_{CBZmax}=9.7720E-7.
The figures are presented in the following order: MEL, 4-NP, and CBZ.





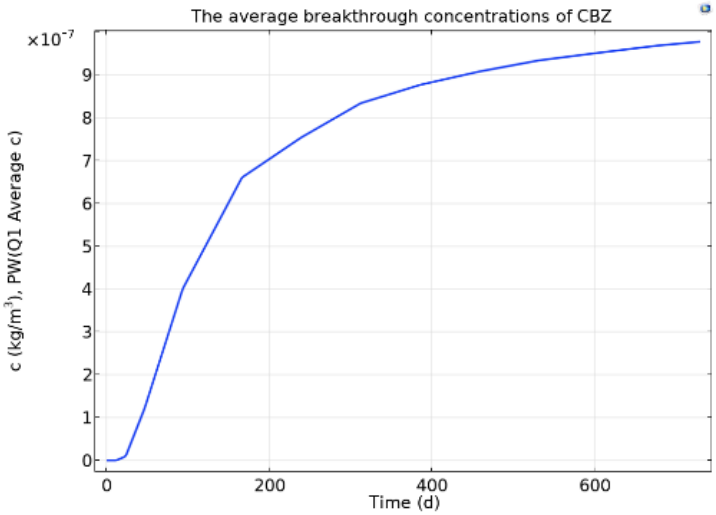


Figure 16 Average breakthrough concentrations of the three PMT substances at the PW under Q_1

Task 17: Effect of Well Distance on Flow and Reactive Transport

Moving the well from 50 m to 150 m away from the river weakens the hydraulic coupling to the river. In the flow plots, In the hydraulic head distribution, the cone of depression interacts less with the river boundary, and the overall head gradient between the river and the well becomes smaller. Consequently, hydraulic travel times increase. The transport plots illustrate that when the well is located 150 m from the river, the contaminant plume advances more slowly toward the well, indicating a delayed breakthrough (larger arrival time and t_{50}). For a conservative tracer, this difference manifests primarily as a time delay, while the final steady-state concentration at the well is expected to remain approximately the same as in the 50 m case.

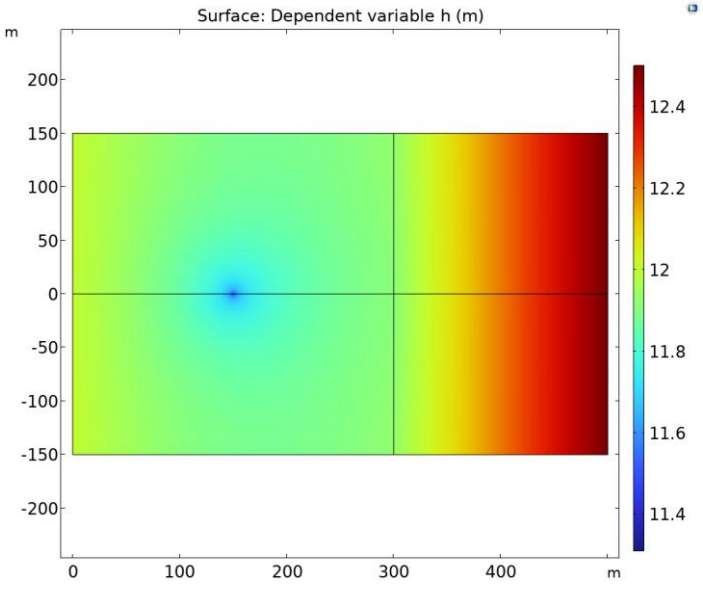


Figure 17 Hydraulic head distribution for well (x=150 m)

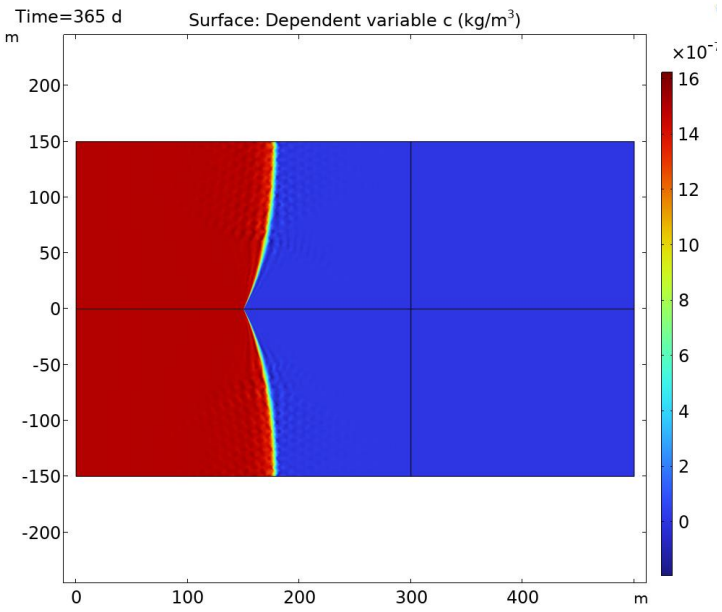


Figure 18 Concentration distribution after 364 days for well (x=150 m)

Task 18: Time-Dependent Concentration Simulation for 4-NP

When a time-dependent 4-NP concentration is applied at the river boundary (linearly interpolated from Table 3), the corresponding plots are shown below. Over the 2-year simulation period, the maximum 4-NP concentration at PW reaches 1.1922×10^{-6} kg/m³ on day 84. Its breakthrough curve is non-monotonic: it first increases, reaches a peak at day 84, and then declines as the boundary source concentration decreases. In contrast, under the constant-concentration scenario, the PW curve rises monotonically and approaches a steady plateau without any subsequent decrease.

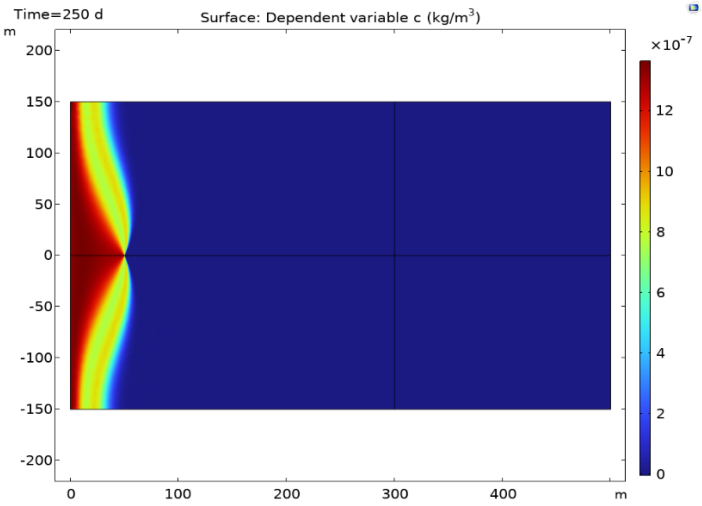


Figure 19 Concentration in the aquifer at t=250 days

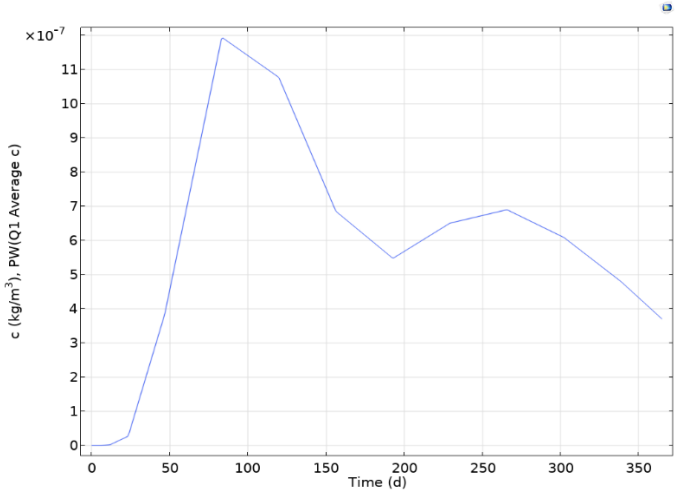


Figure 20 1-D plot of the average concentration at PW over time

Task 19: Evaluation of Model Assumptions and Uncertainties

Three major assumptions and simplifications are:

①. Homogeneous and isotropic aquifer: The model assumes uniform hydraulic conductivity and porosity throughout the domain. In reality, natural aquifers are often heterogeneous, and local variations can significantly affect flow paths and contaminant transport.

②. Simplified boundary conditions: The river boundary and right-hand side are represented by idealized Dirichlet. In reality, river–aquifer interactions are dynamic and may vary seasonally, introducing uncertainty to the recharge and discharge rates.

③. Linear sorption and first-order degradation: The model applies linear equilibrium sorption and first-order decay. However, sorption and degradation often depend on concentration, redox conditions, and microbial activity, leading to nonlinear behavior not captured by the model.

APPENDIX

Task 14

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
q1_path = "D:\A ※ cosmol\Q1 c.csv"
q2_path = "D:\A ※ cosmol\Q2 c.csv"

def load_curve(path):
    for sep in [",", ";", "\t"]:
        try:
            df = pd.read_csv(path, sep=sep, engine="python")
            if df.shape[1] >= 2:
                break
        except Exception:
            df = None
    if df is None or df.empty:
        raise RuntimeError(f"Could not read {path}")

    cols = [c.strip().lower() for c in df.columns]
    df.columns = cols

    time_candidates = [c for c in cols if "time" in c or c == "t"]
    conc_candidates = [c for c in cols if c.startswith("c") or "conc" in c or "concentration" in c]

    time_col = time_candidates[0] if time_candidates else cols[0]
    if conc_candidates:
        conc_col = conc_candidates[0]
    else:
        conc_col = [c for c in cols[1:] if c.startswith("c")] or [cols[1]]
        conc_col = conc_col[0]

    df = df[[time_col, conc_col]].copy().dropna()
    df.columns = ["time_d", "C"]
    df["time_d"] = pd.to_numeric(df["time_d"], errors="coerce")
    df["C"] = pd.to_numeric(df["C"], errors="coerce")
    df = df.dropna().sort_values("time_d")
    df = df.groupby("time_d", as_index=False).mean()
    return df

def compute_t50(df, t_end=365.0):
    times = df["time_d"].to_numpy()
    concs = df["C"].to_numpy()
```

```
    C_end = np.interp(t_end, times, concs)
    target = 0.5 * C_end

    idx = np.where(((concs[:-1] <= target) & (concs[1:] >= target)) |
                   ((concs[:-1] >= target) & (concs[1:] <= target)))[0]

    if len(idx) == 0:
        k = int(np.argmin(np.abs(concs - target)))
        t50 = times[k]
    else:
        k = idx[0]
        t0, t1 = times[k], times[k+1]
        c0, c1 = concs[k], concs[k+1]
        t50 = t0 if c1 == c0 else t0 + (target - c0) * (t1 - t0) / (c1 - c0)

    return dict(t50=t50, t50_round=int(np.round(t50)), C_end=C_end, target=target)

def run_case(path, label):
    df = load_curve(path)
    res = compute_t50(df)
    plt.figure()
    plt.plot(df["time_d"], df["C"], label=f"{label} breakthrough")
    plt.axhline(res["target"], linestyle="--")
    plt.axvline(res["t50"])
    plt.scatter([res["t50"]], [res["target"]])
    plt.xlabel("Time (days)")
    plt.ylabel("Concentration")
    plt.title(f"{label}   t50 ≈ {res['t50_round']} d")
    plt.legend()
    plt.tight_layout()
    plt.show()
    return df, res

df_q1, r1 = run_case(q1_path, "Q1")
df_q2, r2 = run_case(q2_path, "Q2")

print(f"Q1: C_end={r1['C_end']:.6g}, 0.5*C_end={r1['target']:.6g}, t50={r1['t50']:.3f} d -> {r1['t50_round']}
d")
print(f"Q2: C_end={r2['C_end']:.6g}, 0.5*C_end={r2['target']:.6g}, t50={r2['t50']:.3f} d -> {r2['t50_round']}
d")
```

Group Declaration

Task List	Anqi Yu	Qiaowen You	Jiayue Che	Yining Gan
1	√			
2	√			
3	√			
4	√			
5		√		
6		√		
7		√		
8		√		
9		√		
10				√
11				√
12				√
13				√
14				√
15			√	
16			√	
17			√	
18			√	
19			√	
Writing	√	√	√	√
Graphs & Figures	√	√	√	√
Polishing & Quality Control	√	√	√	√
Coffee & cake	√	√	√	√

Declare for Use of AI

Task List	How was Generative AI used? (1, 2, 3, 4) *	Reflections on what was learned using Generative AI
1	1	
2	1	
3	1	
4	1	
5	1	
6	2	Remind me of the relationship between aquifer thickness and drawdown of the well.
7	1	

Task List	How was Generative AI used? (1, 2, 3, 4) *	Reflections on what was learned using Generative AI
8	1	
9	3	Learned how to refine my language.
10	3	Polish report writing.
11	3	Polish report writing.
12	3	Polish report writing.
13	3	Polish report writing.
14	4	Learned how to calculate and plot the time corresponding to half of the maximum pollutant concentration.
15	1	
16	1	
17	1	
18	3	Learned how to refine my language.
19	3	Learned how to refine my language.
Graphs & Figures	1	
Polishing & Quality Control	1	

* Uses of GenAI:
1 No use of GenAI
2 GenAI used for idea generation and structuring the assignment
3 GenAI to edit language/grammar but not content
4 GenAI used to write part of the text/make figures